

Fast and accurate node-level hosting capacity estimation in distribution networks using graph neural networks

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ARTICLE INFO

Keywords:

Graph neural networks

Hosting capacity

Power distribution networks

ABSTRACT

This paper presents a fast and accurate method for estimating node-level hosting capacity in distribution networks using Graph Neural Networks (GNNs). A large open dataset of synthetic networks, representative of Italian distribution grids, was created, and hosting capacity values were computed using AC optimal power flow (AC-OPF) simulations, which are computationally expensive. This dataset was used to train a GNN regressor based on the Graph Isomorphism Network architecture, integrating path-based electrical features to capture both local and global grid dependencies. The proposed GNN is compared to a fast physics-based estimation method, showing a reduction in mean absolute error of about 30%, highlighting its ability to capture the spatial and electrical relationships embedded in the network topology and to produce physically consistent estimates. In terms of efficiency, the model achieves an inference speed over 11,000 times faster than AC-OPF, enabling hosting capacity estimation in milliseconds. This significant speed-up allows real-time applications such as interactive hosting capacity maps, large-scale scenario analysis, and decision support for distribution system operators. The model's robustness is also evaluated under out-of-distribution conditions, where some performance degradation is observed, particularly for small networks, emphasizing the importance of representative training data. Importantly, the GNN remains operational even when traditional power flow methods fail to converge, providing stable and reliable estimates without requiring manual intervention. Overall, the approach demonstrates strong potential for scalable, real-time hosting capacity assessment to support the integration of distributed energy resources.

1. Introduction

The increasing penetration of distributed energy resources (DERs), along with the electrification of key sectors such as transport and heating, has introduced new layers of complexity in the planning and operation of distribution networks. These systems, traditionally designed for passive consumption and unidirectional power flows, must now accommodate dynamic and often unpredictable bidirectional energy exchanges.

In this context, the concept of Hosting Capacity (HC), defined as the maximum amount of generation or load that can be connected without breaching technical constraints, has become a fundamental tool for Distribution System Operators (DSOs). Its relevance stems from the need to manage the expansion and integration of new resources in a decentralized and largely uncontrollable environment. Since DSOs typically

do not control the spatial distribution of new DERs and loads, HC maps, especially when differentiated by technology type, can influence and accelerate connection procedures and support strategic planning for network reinforcement [1].

The topic has been widely investigated in the academic literature, with several reviews and commercial tools already addressing the HC estimation [2–4]. Methodologies vary according to modeling assumptions and computational strategies, but there is broad agreement that the main challenge stems from the nonlinear and nonconvex nature of the power flow equations underlying most formulations [5].

Although nonlinear models are well-established and supported by advanced solvers, they remain computationally intensive, particularly for large-scale systems [2,5,6]. To address this, linear approaches have been proposed that offer a compromise between precision and

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<https://doi.org/10.1016/j.epsr.2026.113871>

Received 14 April 2026; Received in revised form 30 June 2026; Accepted 16 July 2026

Available online 28 July 2026

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computational feasibility [2,5,7–9]. However, even these simplified models can be demanding and may introduce notable errors [10].

1.1. Machine learning for hosting capacity estimation

In recent years, the application of Machine Learning (ML) techniques to HC estimation has gained significant traction within the scientific community [11]. This growing interest is driven by several factors, including the need for scalable methods capable of handling increasingly complex distribution networks, the desire to overcome the computational limitations of traditional optimization-based approaches, and the potential to generalize across diverse network configurations and operating conditions. Moreover, replacing computationally intensive physics-based deterministic methods with ML-based estimations enables real-time and scalable hosting capacity assessment. While simplified approaches based on linearized power flow models and algebraic sensitivity analyses have been proposed to reduce computational burden, these methods typically rely on strong approximations and may fail to capture the non-linear behaviour of distribution networks. As a result, a gap persists between fast but approximate analytical techniques and accurate but computationally intensive simulation-based methods. This shift enables advanced applications such as interactive hosting capacity mapping, large-scale what-if analyses, and real-time decision support for DSOs. In the context of this work, the primary motivation lies in enhancing computational performance (particularly in terms of speed) while achieving an accuracy level that is appropriate for the intended application, acknowledging that acceptable error thresholds may vary depending on the specific use case, and that suitable safety margins can be adopted to account for residual prediction errors.

One of the earliest contributions in this area involved the use of ML for clustering distribution networks based on their HC. This approach allowed DSOs to prioritize detailed analyses on a limited subset of networks, specifically those exhibiting lower performance, thereby facilitating the identification of critical issues [12]. More recent developments have led to the design of neural networks capable of directly estimating HC at the nodal level with very short inference times, making them suitable for real-time applications. For instance, the study presented in [13] emphasizes the importance of fast HC assessments and, in addition to proposing a neural network model, outlines future directions involving the use of Graph Neural Networks (GNNs), supported by preliminary testing.

GNNs offer a distinctive advantage over traditional neural architectures by natively handling graph-structured data [13–15], which aligns naturally with the topology of electrical networks. This capability allows GNNs to model systems of varying size and configuration without retraining, making them particularly suitable for power system applications. Their flexibility and scalability have contributed to their growing adoption in this domain [15,16]. As noted in [17], even when the computational complexity of GNNs is comparable to that of classical optimization models, their use may be preferable in practical settings due to the availability of specialized hardware (such as GPUs) that significantly accelerates neural network inference. A detailed description of the GNN architecture and its mechanism is provided in Section 3.

Another key advantage of ML-based approaches lies in their ability to produce direct estimates of network indicators that are typically obtained through iterative simulations in classical models. Hosting capacity is a prime example, as its estimation often requires numerous power flow or optimization runs. This makes it particularly well-suited for ML-based techniques, which can drastically reduce computation time. Similar benefits have been observed in other applications, such as short-circuit power estimation [18], flexibility potential assessment at the TSO-DSO interface [19], etc. [20] The practical deployment logic of this approach is therefore best understood as an offline–online workflow, detailed in the following subsection.

1.2. Operational use case and deployment rationale

The proposed workflow is summarized in Fig. 1. In the offline phase, a representative set of distribution networks is generated or collected, and node-level hosting capacity values are computed using high-fidelity AC-OPF simulations. Although this step is computationally demanding, it is performed only once to construct the training dataset.

The resulting dataset is then used to train the proposed GNN model, which learns the relationship between network topology, electrical parameters, path-based descriptors, and the corresponding AC-OPF-based hosting capacity values.

After training, the model can be deployed in operational settings. For a new network, hosting capacity estimates are obtained through a single non-iterative forward pass, without solving a new AC-OPF problem for each node. This enables fast assessment of previously unseen networks, provided that their topological and electrical characteristics are reasonably represented in the training data.

In this context, an *unseen network* denotes a network not included in the training set but still belonging to the same statistical domain. Conversely, networks with substantially different size, topology, impedance characteristics, or modelling assumptions should be considered out-of-distribution cases. For such cases, accuracy may degrade, and reliable deployment would require extending the dataset with representative examples followed by retraining or fine-tuning of the GNN.

1.3. Structure of the manuscript

The remainder of this paper is organized as follows. Section 2 introduces the assumptions adopted for hosting capacity estimation and describes the procedure used to generate the synthetic dataset. Section 3 details the architecture of the proposed Graph Neural Network model, including the preprocessing stage and the design of the predictive regressor. Section 4 outlines the experimental setup, training configuration, and evaluation protocol, together with the baseline models used for comparison. Section 5 presents the main results, including prediction accuracy, inference efficiency, learning behavior, and generalization analysis. Finally, Section 6 draws the main conclusions.

2. Hosting capacity assumptions and dataset

2.1. Definition and assumptions of hosting capacity

Hosting capacity has been defined in multiple ways throughout the literature, each tailored to specific applications and computational methodologies [2–4]. In this work, we adopt one of the most widely used definitions: nodal HC, which refers to the maximum amount of generation (or load) that can be individually connected to each node of a distribution network. This indicator has proven to be highly valuable in practical grid management, as demonstrated by its widespread use in HC maps published by DSOs [21]. For the purposes of this work, the analysis is restricted to radial, balanced (single-phase equivalent) distribution networks. More complex configurations, including meshed topologies and unbalanced multi-phase systems, are not considered within the current modelling framework.

One of the main advantages of HC is its ability to provide a direct and intuitive assessment of the network operational state, including the identification of bottlenecks that may hinder the integration of distributed generation or the electrification of demand. The indicator is strongly influenced by network characteristics, topology (which is one of the reasons for investigating GNN models) and load conditions. In particular, when estimating HC for generation, the presence of load tends to increase the available capacity. Consequently, DSOs often adopt conservative assumptions by reducing load levels in order to obtain cautious estimates [21].

In this study, we aim at a simplified preliminary assessment of the suitability of GNNs for HC estimation and therefore neglect both load

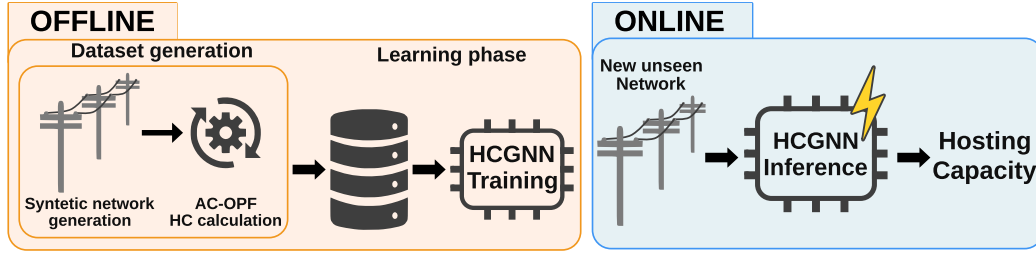


Fig. 1. Offline-online workflow of the proposed HCGNN framework. AC-OPF simulations are used offline to compute node-level hosting capacity values on representative networks and generate the training dataset. Once trained, HCGNN is deployed online to estimate the hosting capacity of new, previously unseen but statistically similar networks through fast, non-iterative GNN inference.

and existing generation. While ignoring the load is a conservative assumption, as it systematically underestimates the actual HC, neglecting existing generation does not share the same conservative nature. More generally, fully neglecting both load and pre-existing generation (including units at non-target nodes) constitutes a notable simplification with respect to real-world operating conditions. Nevertheless, the impact of pre-existing generation can be readily incorporated a posteriori by subtracting its contribution from the HC estimated by the GNN. For practical deployment in live networks, however, the proposed model would require either dedicated post-processing steps or retraining to dynamically account for variations in load and installed generation.

Importantly, these simplifications do not affect the validity of the model accuracy evaluation, since both training and validation datasets are generated under the same assumptions. In this context, the primary objective of this study is not to assess the impact of such simplifications on the absolute value of the hosting capacity, but rather to investigate the capability of ML models to accurately learn and reproduce the outcomes of physics-based computational procedures, independently of the specific modelling assumptions adopted.

2.2. Dataset generation

Hosting Capacity is accurately determined through a sequence of AC-OPF computations. For each node in the network, a dedicated optimization problem is formulated to maximize the power injection at that specific node, subject to the network operational constraints. These constraints can vary widely [2,3], but in this study (as in most practical cases) we consider two primary types: voltage limits, which are often among the most restrictive [2,3], and thermal limits on lines and transformers.

Classical AC-OPF formulations typically compute the hosting capacity by solving an optimization problem that directly maximizes the power injected by a fictitious generator connected at node n :

$$\max P_{g(n)}. \quad (1)$$

The maximization is subject to the selected network constraints. The resulting optimal value of $P_{g(n)}$ therefore represents the hosting capacity of node n .

The problem is inherently complex and computationally demanding. The formulation of AC-OPF is notoriously nonlinear and nonconvex and must be solved repeatedly for each node n in the network. This computational burden remains significant even under the simplified conditions adopted in this study, including the single-phase representation, the absence of load, and the reduced set of constraints. These considerations are further confirmed by the computational times reported in Section 5.2 and by the convergence difficulties encountered during the experimental phases, as discussed in Section 5.4.

To train and validate the proposed GNN model, a dataset of 7188 synthetic distribution networks was generated using the methodology described in [22]; the resulting networks are collected in the repository [23]. The network generation algorithm was configured to produce topological and electrical characteristics representative of typical

Table 1

Summary of the generated dataset.

Category	Feature	Value
Dataset size	Number of networks	7188
	Total nodes and edges	1.43×10^6
Network size	Nodes per network	68-559 (avg. 199)
	Topology	Radial
Electrical parameters	Resistance r [p.u.]	$2.5 \times 10^{-5} - 7.5 \times 10^{-3}$
	Reactance x [p.u.]	$1.3 \times 10^{-5} - 1.1 \times 10^{-2}$
	Max loading S_{max} [p.u.]	$3.41 - 17.97$
Target variable	Hosting Capacity [p.u.]	$0.56 - 10.0$ (avg. 4.27)

medium-voltage Italian distribution systems. The HC was then evaluated by considering the thermal ratings of electrical lines as limiting factor, together with a voltage constraint of $\pm 5\%$.

Additionally, to better capture the nonlinear behavior of the networks, the dataset includes configurations with deliberately extended line lengths (twice the standard planning guidelines), thus enhancing the model robustness in handling higher impedance values and low R/X ratios, which may occasionally occur in real-world scenarios. For each network and for every node within it, the hosting capacity was computed through dedicated AC-OPF simulations, resulting in a comprehensive dataset suitable for training and validating the GNN.

While the generation of this dataset requires a significant computational effort, it is performed only once. The resulting trained model can then be deployed to provide instant HC estimates without the need for further OPF simulations.

The main characteristics of the generated dataset are summarized in Table 1. The dataset exhibits substantial variability in network size, with feeders ranging from 68 to 559 nodes (average 199), ensuring a wide spectrum of structural complexity. All networks are connected and strictly radial, with an average node degree close to 2, consistent with typical distribution feeders.

Electrical parameters span realistic ranges, with both resistance and reactance covering several orders of magnitude, and line thermal limits varying between 3.41 and 17.97 MVA. Hosting capacity values are well distributed across nodes, ranging from 0.56 to 10.0 (mean 4.27), providing a sufficiently rich target space for supervised learning.

Overall, the dataset combines structural diversity and physically consistent electrical properties, supporting robust training and reliable evaluation of the proposed model.

3. The architecture

3.1. Background on graph neural networks

Graph Neural Networks [24,25] are a class of machine learning models specifically designed to process graph-structured data composed of nodes and edges. In power systems, this representation arises naturally, as buses correspond to nodes and electrical lines or transformers to

edges [26,27]. GNNs operate by allowing each node to exchange information with its neighbors, progressively building a latent representation that captures both local properties and the surrounding network context [28,29].

At each layer, node features are updated through a message-passing mechanism that aggregates information from neighboring nodes and combines it with the node's own state [28]. By stacking multiple layers, information can propagate over increasingly larger portions of the network, enabling each node to encode structural and electrical characteristics beyond its immediate neighborhood [29].

Among the various GNN architectures proposed in the literature, this work adopts the Graph Isomorphism Network (GIN) [30] due to its high expressive power in distinguishing graph structures. This property is particularly relevant for hosting capacity estimation, since nodes with similar electrical parameters may have different capacity limits depending on their position in the feeder, branching structure, and upstream connectivity [4–6,9,10,21]. In each GIN layer, neighboring node representations are summed and combined with the node's own features through a multilayer perceptron. The tunable self-contribution parameter allows the model to balance node-specific electrical descriptors with aggregated neighborhood information, improving its ability to capture structural differences in radial distribution grids.

It is important to distinguish between the training and inference phases of the proposed GNN-based approach. Training is performed offline and involves an iterative optimization process, common to all neural network models, in which the network parameters are learned from data over multiple epochs [31]. Once training is completed, the resulting model has fixed parameters and architecture. During inference, which represents the operational use case of interest, hosting capacity estimates are obtained through a single forward pass with a predefined number of message-passing layers. As a result, inference does not require any iterative procedure or convergence process, and its computational cost is deterministic and well suited for real-time and large-scale applications.

Although GNNs are effective at modeling local interactions, hosting capacity depends not only on neighboring nodes but also on the position of a node within the feeder and on cumulative electrical effects along upstream paths. This limitation is related to the difficulty of propagating long-range information in standard message-passing GNNs, often discussed in terms of oversquashing [32,33]. While graph transformer architectures can capture long-range dependencies through global attention mechanisms [34–36], they often entail higher computational costs and reduced inductive bias for power system applications. To address this limitation while preserving efficiency, the next subsection introduces a preprocessing stage that enriches each node with path-based electrical features, enabling the proposed model to capture long-range dependencies without requiring deep or computationally expensive architectures.

3.2. Preprocessing

To provide the GNN with a richer representation, we augment each node with features derived from its relative position within the feeder and from the electrical properties of upstream lines. Specifically, the feeder node, corresponding to the upstream supply point, is selected as the root of the graph. For every other node, the shortest path to the feeder is identified, and summary statistics of the edge attributes along this path are computed.

For each node, we extract the following features:

- *structural descriptors*: node degree (number of directly connected lines), depth and normalized depth (distance from the feeder in terms of network layers), leaf indicator (identifying terminal buses), closeness centrality (reflecting how electrically close a node is to the rest of the network), and average neighbor degree (capturing local connectivity patterns);

- *path statistics*: minimum, maximum, mean, and cumulative (sum) values of line resistance r and reactance x , together with minimum, maximum, and mean values of the maximum loading capacity S_{max} , are computed along the path connecting each node to the feeder. These features summarize both local and cumulative electrical constraints affecting power injection at that location.

This preprocessing stage is a key component of the proposed hybrid architecture. Rather than forcing the GNN to learn long-range feeder dependencies exclusively through repeated local message passing, the model is provided with explicit path-based electrical summaries from each node to the feeder. These descriptors encode cumulative impedance and limiting-capacity information along the upstream path, which are directly related to voltage drops and thermal constraints affecting hosting capacity. As a result, the subsequent GIN layers can focus on refining local graph representations, while the global electrical context is already available at the node-feature level. This design mitigates the need for very deep GNN architectures, reduces the impact of *oversquashing*, and preserves the low computational cost required for real-time inference.

3.3. Hosting capacity GNN (HCGNN)

The predictive model, referred to as **HCGNN**, is a node-level regressor composed of stacked GIN layers followed by an output MLP (multilayer perceptron), a design consistent with recent applications of GNNs to regression tasks on graph-structured data [37].

Each GIN layer is implemented as a GINConv block, i.e., a *graph convolutional operator that updates each node representation by aggregating the features of its neighboring nodes and combining them with the node's own features through a learnable nonlinear function*. In this work, the update function is implemented as a four-layer MLP with nonlinear activation. Batch normalization, ReLU, and dropout are applied after each convolution to improve training stability and reduce overfitting. This design is particularly effective given the heterogeneous structure of the input distribution networks.

After the last GIN layer, node embeddings are passed through an MLP consisting of $(\ell - 1)$ hidden layers of size h , each followed by batch normalization, activation, and dropout, and a final linear layer that outputs a scalar prediction.

4. Experimental setting

4.1. Training setup

All experiments were conducted on an Intel® Core™ i9-14900HX CPU (32 GB RAM) and an NVIDIA RTX 4070 Max-Q GPU (8 GB VRAM), using PyTorch Geometric [38] with CUDA acceleration. Each experiment was repeated five times with different random seeds to ensure reproducibility. Before every run, the random number generators of Python, NumPy [39], and PyTorch [40] were re-initialized, and deterministic execution was enforced by disabling non-deterministic CUDA kernels.

The entire dataset was constructed using the AC-OPF Pandapower library [41] to compute the HC for every node in each synthetic network. The dataset was randomly shuffled and divided into 70% for training, 15% for validation, and 15% for testing. This partition was fixed across all runs.

4.2. Optimization and training

The model was trained using the Adam optimizer with batch size 512, and Smooth L1 (Huber) loss with $\beta = 2.0$, chosen for its robustness to outliers. Training was performed for a maximum of 1500 epochs, with early stopping based on the validation loss and a patience of 20 epochs. The best-performing model on the validation set was restored for final evaluation.

Table 2

Final hyperparameter configuration of the proposed GIN model.

Hyperparameter	Value
Number of GIN layers	3
Hidden dimension	128
MLP layers in each GIN block	4
Activation function	ReLU
Dropout rate	0.1
Batch normalization	Yes
Batch size	512
Optimizer	Adam
Learning rate	10^{-2}
Loss function	Smooth L1, $\beta = 2.0$
Early stopping patience	20 epochs
Maximum number of epochs	1500

The final configuration was selected through grid search over the following discrete values: number of GIN layers in {2, 3, 4, 5}, hidden dimensions in {64, 128, 256}, dropout rates in {0.0, 0.1, 0.2, 0.3}, and learning rates in $\{10^{-3}, 10^{-2}\}$. The selected architecture consists of 3 GIN layers, 128 hidden units, 4-layer MLP aggregators, ReLU activation, Batch Normalization applied to the hidden node embeddings in each convolution block, and dropout $p = .1$. The main hyperparameters used in the final model are summarized in Table 2.

4.3. Evaluation protocol

Model performance was evaluated on the test set using the Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). Both metrics were averaged over five independent runs to account for variability. They are defined as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad \text{MAPE} = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad (2)$$

where y_i and $\hat{y}_i$ denote the true and predicted hosting capacity for node i , and N is the total number of samples.

4.4. Baseline models

To assess the effectiveness of HCGNN architecture, its performance was compared against a recent method for hosting capacity estimation based on the direct inversion of a linearized power flow model [42], referred to here as LINEAR DISTFLOW. In this formulation, each generic branch ij is subject to the following voltage-drop constraint:

$$W_i - W_j = 2r_{ij}P_{ij} + 2x_{ij}Q_{ij}, \quad (3)$$

where W_i and W_j denote the squared voltage magnitudes at buses i and j respectively, r_{ij} and x_{ij} are the branch resistance and reactance, while P_{ij} and Q_{ij} are the active and reactive power flows. These flows must also satisfy the thermal limit:

$$P_{ij}^2 + Q_{ij}^2 \leq S_{ij(max)}^2, \quad (4)$$

with $S_{ij(max)}$ being the rated power capacity of branch ij .

To complete the model, nodal power balance equations (standard in any power flow formulation) are included for both active and reactive power, ensuring that the net injection at each bus equals the sum of incoming and outgoing flows. Together with the voltage-drop and thermal limit constraints, these equations define the core structure of a linearized power flow model used for hosting capacity estimation.

For a generic radial distribution network with N buses and $L = N - 1$ branches, the model comprises:

- L voltage-drop equations,
- L thermal limit constraints,
- $2N$ nodal active and reactive power balance equations.

Table 3

Comparison of prediction errors between the baseline (LINEAR DISTFLOW) and the proposed GNN.

Model	MAE [p.u.]	MAPE [%]
LINEAR DISTFLOW	0.1010	2.73
HCGNN	0.0705 ± 0.0083	2.17 ± 0.23

Table 4

Tolerance-aware feasibility rate of HCGNN with different prevailing limit (voltage/thermal).

κ	1.00	1.01	1.02	1.03	1.04	1.05
FR(κ) - thermal	0.56	0.99	1.00	1.00	1.00	1.00
FR(κ) - voltage	0.71	0.75	0.79	0.82	0.84	0.86
FR(κ) - all	0.60	0.93	0.95	0.95	0.96	0.97

Once this system of equations is constructed, it can be directly inverted to compute, for each network node, the maximum active power injection that either causes a voltage magnitude violation at any bus or leads a branch to reach its rated power limit [43].

LINEAR DISTFLOW offers a good balance between accuracy and computational speed. Although it retains the iterative, node-by-node nature typical of AC-OPF-based formulations, it requires only a single matrix inversion, significantly reducing computational overhead. Nonetheless, it can still be computationally demanding for large-scale systems, as its complexity grows polynomially with network size, impacting on its scalability.

Despite this, LINEAR DISTFLOW remains a robust and reliable choice for practical applications requiring accurate and reasonably fast hosting capacity assessments. Its combination of simplicity, speed, and precision establishes it as a good reference method, making it particularly well-suited as a benchmark for evaluating HCGNN computational performance.

5. Results

5.1. Prediction performance

Table 3 reports the test-set results of HCGNN compared with LINEAR DISTFLOW. Values represent the mean $\pm$ standard deviation over five runs.

The GNN reduces the mean absolute error by approximately 30% and the relative error by about 20% compared to the baseline. These gains indicate that the model effectively captures the spatial and electrical dependencies encoded in the grid topology, yielding more consistent and physically plausible hosting capacity estimates across different networks. The small standard deviations also suggest stable training dynamics and good generalization.

To further assess whether the predicted hosting capacity values are conservative, we compute a feasibility rate defined as

$$\text{FR}(\kappa) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\hat{y}_i \leq \kappa y_i), \quad (5)$$

where y_i is the ground-truth hosting capacity, $\hat{y}_i$ is the prediction of HCGNN, and $\kappa \geq 1$ controls the allowed tolerance with respect to the true value. The case $\kappa = 1$ corresponds to strict feasibility, i.e., predictions that do not overestimate the ground truth. The results of this analysis are reported in Table 4 for different prevailing limiting constraints.

Under the strict feasibility criterion, HCGNN produces conservative predictions for 60% of the test samples. This value is expected, since the model is trained to minimize prediction error and is not explicitly constrained to penalize overestimation more strongly than underestimation. However, the feasibility rate increases sharply when a small tolerance is allowed: with only 1% tolerance, 93% of the predictions satisfy $\hat{y}_i \leq 1.01y_i$, and with 5% tolerance the rate reaches 97%. This

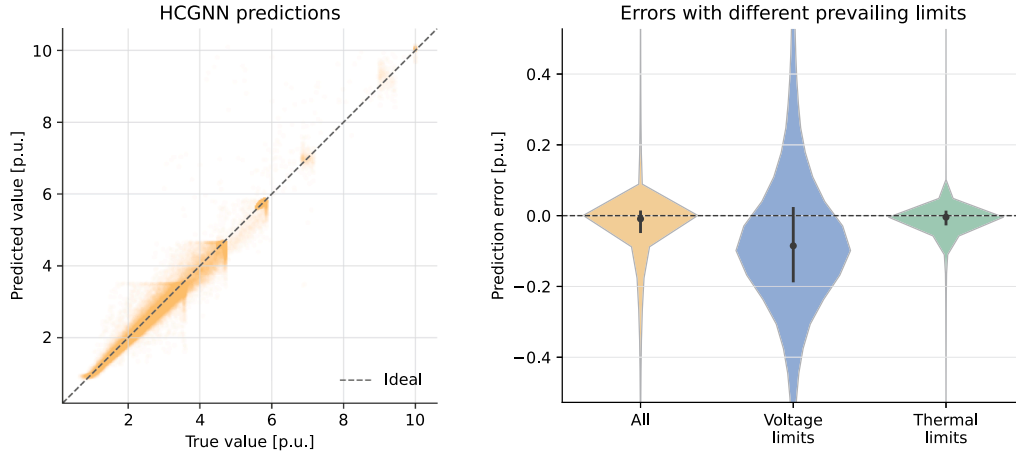


Fig. 2. Predicted versus true hosting capacity values and signed prediction errors for HCGNN. In the left panel, the dashed line denotes the ideal prediction $\hat{y} = y$; points below the line correspond to conservative predictions, whereas points above it indicate overestimation. The right panel shows the distribution of prediction errors $\hat{y} - y$ over all test cases and over the subsets dominated by voltage and thermal limits, with the dashed horizontal line marking zero error.

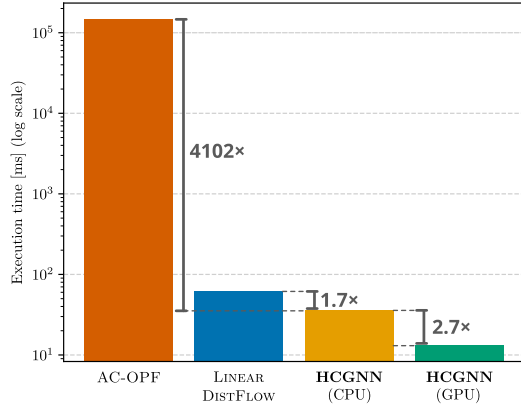


Fig. 3. Comparison of average execution times (log scale) for hosting capacity computation using the full AC-OPF simulator, the LINEAR DISTFLOW baseline, and the proposed HCGNN model (CPU/GPU). The GNN achieves over four orders of magnitude speed-up compared to the simulator and remains substantially faster than the linear method.

suggests that most infeasible predictions are only marginally above the true hosting capacity.

A more detailed analysis highlights a different behaviour depending on the HC limiting constraint, which can be readily identified by analysing the AC-OPF results with respect to their corresponding operational limits (i.e., voltage magnitude and thermal loading). In particular, thermal constraints exhibit a very high feasibility rate even under small tolerances, rapidly reaching near-perfect values already for $\kappa \geq 1.02$. This indicates that the model is generally able to accurately capture thermal limits and rarely produces significant overestimations in thermally constrained conditions. On the contrary, voltage constraints appear more challenging, with a lower feasibility rate and a more gradual increase as the tolerance parameter grows. This behaviour is consistent with the higher sensitivity of voltage profiles to network conditions and power injections, as well as to the non-linear nature of power flow equations, which are inherently more difficult to approximate through data-driven models. This behaviour is also reflected in the error distribution, where voltage-related limitations emerge as the predominant source of inaccuracies, consistent with the trends shown in Fig. 2, which reports the scatter plot of the predicted versus true values (left) and the corresponding statistical distribution of the error (right), where predictions are tightly clustered around the ideal line with only minor deviations above the diagonal.

These results also provide a strong motivation for future physics-informed GNN formulations, in which feasibility constraints, power-flow residuals, or asymmetric penalties on overestimation could be incorporated directly into the training objective. It is worth noting that voltage-related constraints are active in only approximately 25% of the samples in the considered dataset, which inherently limits the model's ability to accurately learn and generalize this phenomenon. This imbalance in the training data may partially explain the lower feasibility rates observed for voltage-constrained cases. In this respect, targeted dataset construction or sampling strategies, aimed at increasing the representation of voltage-critical conditions, could further improve the model's performance in capturing voltage-related limitations.

5.2. Inference efficiency

The computational performance of the proposed model was assessed in terms of inference time for hosting capacity estimation. Fig. 3 compares the average execution time of the full nonlinear AC-OPF simulator, the LINEAR DISTFLOW baseline, and the proposed HCGNN executed on both CPU and GPU. While the AC-OPF remains the most accurate reference, it is computationally prohibitive, over four orders of magnitude slower than the neural surrogate.

The LINEAR DISTFLOW formulation already provides a considerable reduction in computational cost compared to the nonlinear simulator.

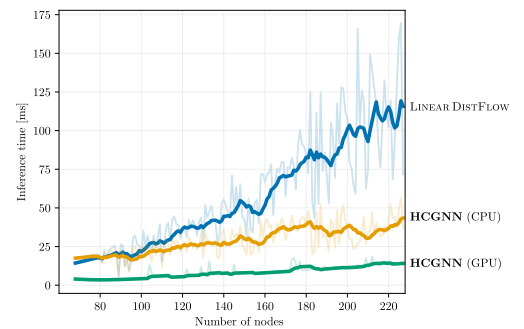


Fig. 4. Inference time versus network size. The linear baseline is faster only for very small grids, after which the GNN consistently outperforms it. The GPU implementation offers the best scalability and enables real-time evaluation. Thin/light lines represent actual inference times for individual networks, while thick/dark lines show the average values obtained by aggregating results for networks of similar size.

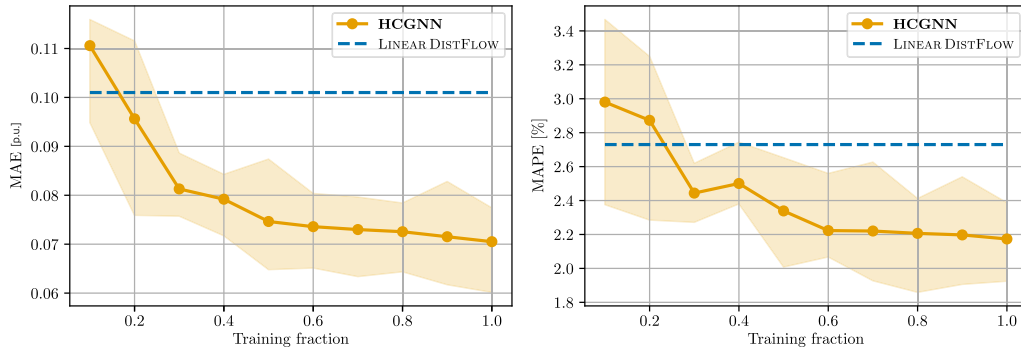


Fig. 5. Learning curves of MAE and MAPE as a function of training set size (where a fraction of 1.0 corresponds to 70% of the available dataset). The proposed HCGNN quickly surpasses the LINEAR DISTFLOW baseline once sufficient data are available, achieving lower errors and stable convergence.

Table 5
Performance comparison on the IEEE 123-bus benchmark.

Model	MAE [p.u.]	MAPE [%]
LINEAR DISTFLOW	0.340	10.903
HCGNN	0.117 $\pm$ 0.025	5.57 $\pm$ 1.37

However, the proposed HCGNN achieves an even greater improvement in efficiency: it is about 1.7 $\times$ faster than the linear model on CPU, and the GPU implementation further accelerates inference by 2.7 $\times$, yielding a total speed-up of approximately 4.6 $\times$ over the linear baseline. When compared to the full AC-OPF simulator, the gain exceeds four orders of magnitude ($\approx 11000\times$ faster), reducing computation from minutes to just a few milliseconds per network instance.

Fig. 4 further analyzes the inference time as a function of network size. The linear baseline exhibits lower latency only for very small systems (fewer than about 90 nodes), after which the GNN consistently outperforms it due to its fast forward pass. The GPU implementation of the HCGNN is the fastest across all cases, maintaining nearly flat scaling with respect to the number of nodes. The AC-OPF simulator is omitted from this figure for clarity, as its runtime is several orders of magnitude higher (see Fig. 3). Overall, these results confirm that the proposed model combines high predictive accuracy with exceptional computational efficiency, enabling real-time and scalable hosting capacity assessments.

5.3. Learning behavior

Fig. 5 shows the evolution of the prediction error as a function of the training set size. Both the MAE and the MAPE decrease monotonically as more data are used for training, indicating stable and data-efficient learning dynamics. For very limited training data (below roughly 20% for MAE and 25% for MAPE), the performance of the proposed HCGNN is slightly inferior to the LINEAR DISTFLOW baseline, reflecting the higher data requirements of neural models. However, as the training fraction increases, HCGNN rapidly surpasses the LINEAR DISTFLOW approach and achieves substantially lower errors, demonstrating its superior capacity to capture the nonlinear relationships that govern hosting capacity. The shaded regions represent the range between the minimum and maximum error observed across the five random seeds, highlighting the consistency of the results and the robustness of the training process.

5.4. Convergence robustness on the IEEE 123-bus system

Another practical advantage of the proposed HCGNN model over the traditional simulator lies in its robustness to convergence issues. When applying the full AC-OPF to the IEEE 123-bus distribution feeder

Table 6
Performance comparison on the IEEE 33-bus benchmark.

Model	MAE [p.u.]	MAPE [%]
LINEAR DISTFLOW	0.014	3.49
HCGNN	0.204 $\pm$ 0.052	11.191 $\pm$ 1.02

[44], several nodes fail to yield a valid hosting capacity due to numerical non-convergence of the optimization solver. It is worth noting that these issues are not related to problem infeasibility, but rather to numerical difficulties that can typically be resolved through fine-tuning of the solver convergence parameters. However, this process requires manual user intervention and is therefore time-consuming. In contrast, HCGNN produces consistent and physically plausible estimates for all network nodes (see Table 5), regardless of operating conditions or initialization.

Fig. 6 illustrates this aspect: nodes for which the AC-OPF fails to compute a result are highlighted with red circles. The labels pointing to these nodes indicate the hosting capacity value predicted by HCGNN, demonstrating that the proposed model can seamlessly provide estimates even in cases where conventional power flow formulations break down. This property further emphasizes the practical applicability of data-driven approaches for large-scale and real-time hosting capacity analysis, eliminating the need for manual intervention or solver tuning.

5.5. Generalization under out-of-distribution conditions: IEEE 33-bus system

To further evaluate the generalization capability of the proposed model, an additional experiment was conducted on the well-known IEEE 33-bus radial distribution network, a standard benchmark widely used in power flow studies [45]. Consistently with the assumptions adopted throughout this work, all existing loads and generators were excluded from the network. Voltage limits were set to the commonly adopted $\pm 5\%$ of the nominal value, and all lines were assigned a uniform thermal rating of 6 MVA. This setup enables a direct comparison with analytical methods on a topologically consistent system.

It is important to note that the HCGNN model was trained exclusively on the synthetic networks described in Section 2, which comprise medium- and large-scale topologies. Therefore, testing on the much smaller IEEE 33-bus network provides an out-of-distribution (OOD) scenario, allowing us to sample the robustness of the model under significant domain shift.

Table 6 summarizes the results obtained by the proposed HCGNN and the LINEAR DISTFLOW baseline on this benchmark. Each result is averaged over five independent runs, with the corresponding standard deviation reported.

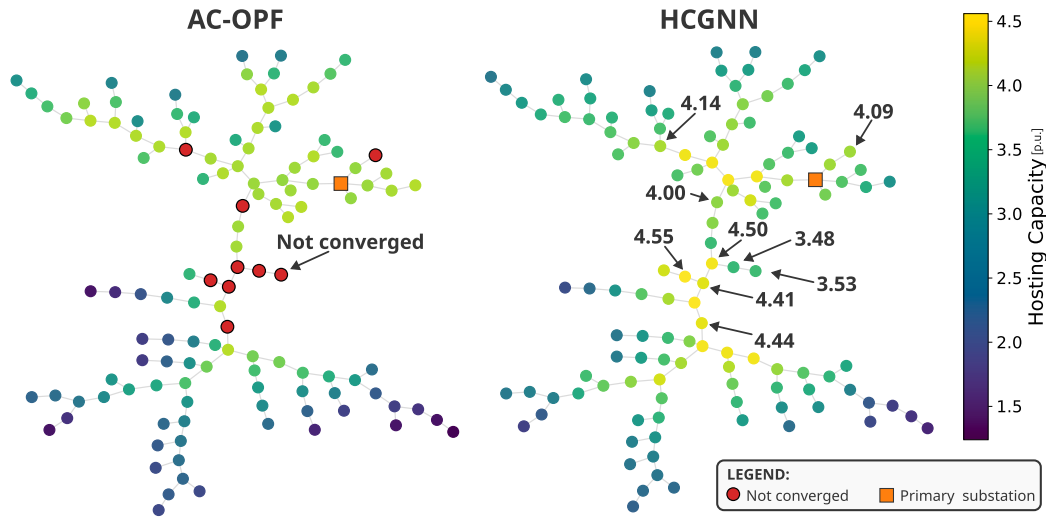


Fig. 6. Comparison between AC-OPF and HCGNN on the IEEE 123-bus system. Nodes where the AC-OPF simulator fails to converge are highlighted in red (left). On the right, HCGNN provides hosting capacity estimates for all nodes, including those where AC-OPF does not converge, with numerical values (in [p.u.]) indicated and a color scale representing the available capacity.

As shown in Table 6, HCGNN exhibits significantly higher prediction errors than the linear baseline on the IEEE 33-bus system. This result confirms that the model has restricted out-of-distribution generalization capability when applied to networks that are substantially different from those used for training.

This degradation is expected. The training dataset described in Section 2 consists of medium- and large-scale synthetic feeders, with network sizes ranging from 68 to 559 nodes. The IEEE 33-bus system is therefore outside the size range observed during training and also differs in terms of compactness and electrical characteristics. As a result, the GNN is required to extrapolate beyond the statistical domain on which it was trained.

Importantly, this limitation does not indicate a structural inability of the proposed approach to handle small networks. Rather, it highlights the need for representative training data. In practical deployments, this issue could be addressed by extending the training dataset to include smaller feeders, for instance networks in the 20–50 node range, followed by retraining or fine-tuning of the GNN. In this work, the IEEE 33-bus case is intentionally used as an out-of-distribution test to quantify the effect of a strong domain shift.

6. Conclusion

This paper presented a GNN approach for fast and accurate hosting capacity estimation in distribution networks. The proposed model, referred to as HCGNN, was trained on a large collection of synthetic feeders representative of typical Italian medium-voltage systems. By leveraging enriched node features and the expressive power of the GIN architecture, the model effectively captured both local and global electrical dependencies.

Experimental results demonstrated that the HCGNN significantly outperforms a state-of-the-art linearized power flow formulation in terms of prediction accuracy, while achieving computational speed-ups of several orders of magnitude compared to full AC-OPF simulations. The GPU implementation enables real-time inference, highlighting the potential of data-driven models as practical tools for DSOs.

Importantly, the proposed GNN is not intended to replace AC-OPF-based hosting capacity calculations in high-fidelity offline studies, but rather to complement them by enabling fast and robust assessments in operational contexts where repeated or real-time evaluations are required. While AC-OPF remains essential for generating accurate reference solutions and training data, its computational burden and conver-

gence issues limit its applicability in large-scale or interactive use cases, which are effectively addressed by the proposed approach.

Overall, the proposed framework provides a scalable and reliable solution for rapid hosting capacity assessment, supporting the efficient integration of distributed energy resources into modern power grids.

Future works. Future work will focus on improving the generalization capabilities of the proposed GNN, particularly under out-of-distribution conditions. In addition, the modelling framework can be extended to incorporate load and generation profiles within distribution networks, as well as more complex grid configurations, including meshed topologies and unbalanced multi-phase systems. Another research direction involves the development of temporal GNN models capable of capturing both the evolution of network topology and temporal load profiles. Such models could enable the prediction of additional performance indicators, for example flexibility needs, broadening the applicability of the proposed approach. Finally, future work will also investigate explainability methods for GNNs, such as feature-attribution and graph-based explanation techniques, to improve the interpretability of the proposed data-driven model and support its practical adoption by DSOs.

CRedit authorship contribution statement

Antonio Longa: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization; **Marco Rossi:** Writing – original draft, Visualization, Software, Investigation, Formal analysis, Data curation; **Francesca Soldan:** Writing – original draft, Visualization, Validation, Formal analysis, Conceptualization; **Ramon Zambetti:** Writing – original draft, Software, Data curation; **Andrea Cazzaniga:** Writing – original draft, Supervision, Project administration, Data curation, Conceptualization; **Andrea Passerini:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Microsoft Copilot to assist with language editing. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work has been financed by the Research Fund for the Italian Electrical System under the Three-Year Research Plan 2025-2027 (MASE, Decree n.388 of November 6th, 2024), in compliance with the Decree of April 12th, 2024, and by the Research Council of Norway through its Centre of Excellence Integreat - The Norwegian Centre for knowledge-driven machine learning, project number 332645.

Data availability

The public data repository is available at [23].

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